MACHINE LEARNING WITH SCIKIT-LEARN

Machine Learning with scikit-learn

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Course overview — what you will learn, how the course works, and how to use AI assistants well.

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Master's Degree Program · Hands-on with Google Colab

AGENDA

Today's agenda

01 **What this course is about**
From data to a working model — the practical path

02 **Who this course is for**
Prerequisites and mindset

03 **Course roadmap**
8 sessions, 8 labs, 1 final project

04 **How each session works**
Slides → demo → lab in Colab

05 **Your toolkit**
Google Colab · scikit-learn · pandas · seaborn

06 **Vibe coding policy**
Using AI assistants without fooling yourself

THE BIG PICTURE

What this course is about

You will learn to turn raw data into models that predict something useful — and to prove they actually work.

8

hands-on sessions
slides + Colab lab every time

10+

models you can use
and explain the next day

1

end-to-end capstone
churn prediction project

PREREQUISITES

Who this course is for

You already have

Some Python — variables, loops, functions, maybe pandas. You can read code and modify it.

You do NOT need

A math degree. We keep equations light: intuition first, formulas only when they earn their place.

You will gain

A working ML workflow: clean data, build pipelines, pick models, evaluate honestly, tune, ship.

Mindset: run everything yourself. In this course, the lab notebook matters more than the slides.

ROADMAP

8 sessions — fundamentals to project

1
01 · Intro to ML
first model in 5 lines

→

2
02 · Data cleaning
understand & tidy data

→

3
03 · Pipelines
features & leakage

→

4
04 · Evaluation
train/test + CV + metrics

1
05 · Regression
Linear → trees

→

2
06 · Classification
+ ensembles

→

3
07 · Unsupervised
clustering, PCA + tuning

→

4
08 · Capstone
churn project

Sessions 1–4: the foundations every model needs. Sessions 5–8: model families, then a full project.

RHYTHM

How each session works

Slides (~60 min)
concepts in English

→

Demo notebook
run along, same code as slides

→

Lab notebook (~60 min)
โจทย์ภาษาไทย in Colab

→

Solution
unlocked after the lab

Every file lives in one repository

slides/ (PDF + PPTX) · demos/ (run-along notebooks) · excercises/ (Thai labs) · excercises/solutions/ (worked answers). Open any notebook in Colab with one click — no installs, no downloads.

Assessment

Labs 40% (complete + honest attempts) · Final project 40% (working pipeline + honest evaluation + short presentation) · Participation 20% (demo follow-along, questions).

TOOLKIT

Your toolkit — already installed in Colab

Google Colab

Free Jupyter in the browser. GPU not needed for this course. Your work saves to Google Drive.

scikit-learn

The ML library. One consistent API: fit() → predict(). Classification, regression, clustering, preprocessing, tuning.

pandas

Tables (DataFrames): load, inspect, clean, transform data. The first tool you touch in every project.

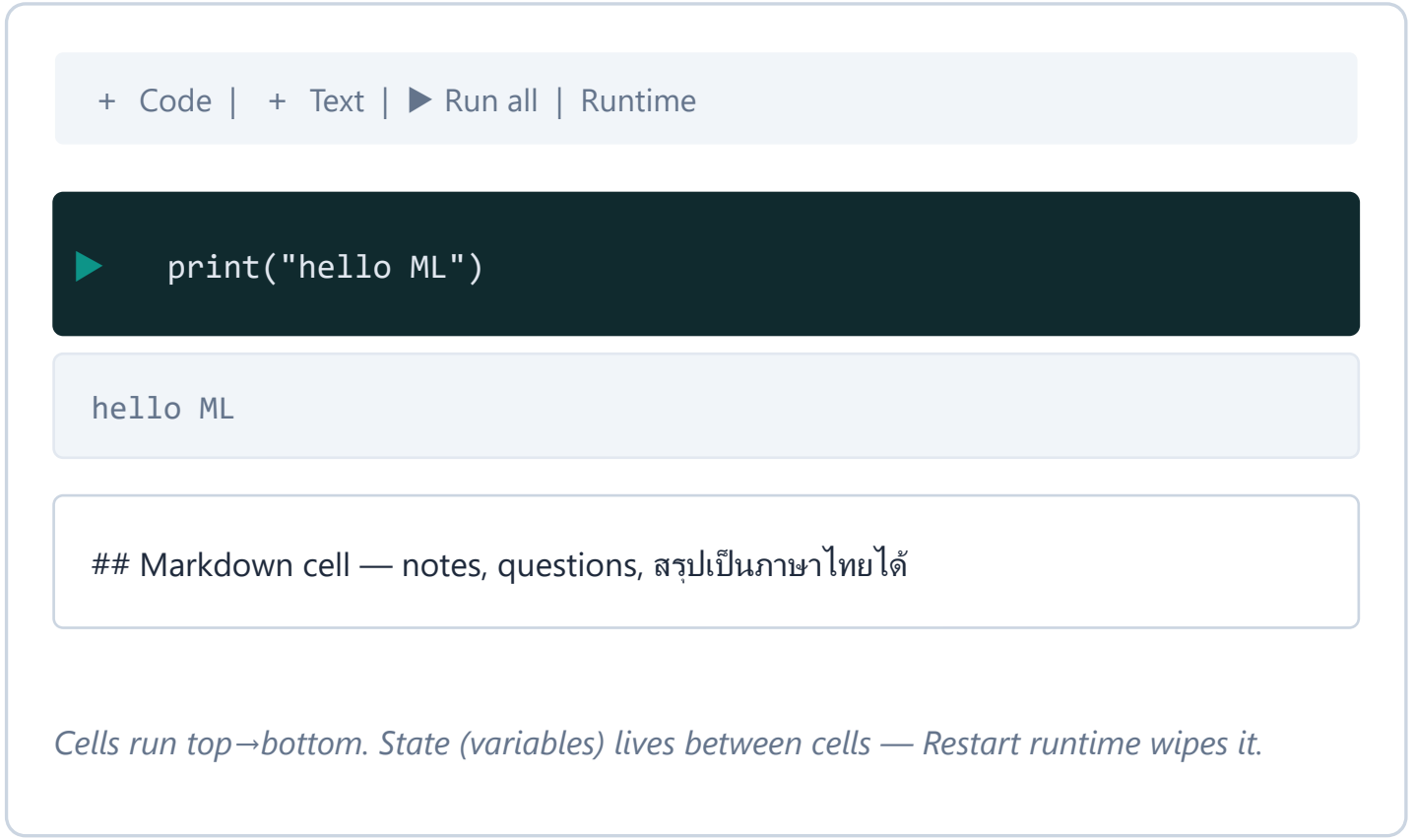
seaborn + matplotlib

Plots: distributions, correlations, decision boundaries. Seeing your data comes before modeling it.

```
# Everything above is pre-installed in Colab. Just import and go:
import pandas as pd                    # tables
import seaborn as sns                  # plots
from sklearn.model_selection import train_test_split   # the ML library
```

COLAB IN 60 SECONDS

Google Colab — your lab bench



- **Shift + Enter runs the current cell**
- Everything is in the cloud — nothing to install
- File → Save a copy in Drive keeps your work
- Runtime → Run all = “fresh run” sanity check
  - Number on a cell = the order it ran

REFRESHER

The 20% of pandas you'll use 80% of the time

```
df = pd.read_csv("titanic.csv")        # load a table
df.head()                              # first 5 rows
df.info()                              # columns, types, missing
df.describe()                          # quick statistics
df["age"]                              # one column (Series)
df[["age", "fare"]]                     # two columns
df[df["age"] > 30]                     # filter rows
df.groupby("sex")["survived"].mean()    # group + aggregate
df["fare"].fillna(df["fare"].median())  # fill missing
df["fare"].isna().sum()                # count missing
df.sort_values("fare").tail(3)         # biggest fares
df["sex"].value_counts(normalize=True) # category shares
```

- **A DataFrame is a table**
- columns = Series (one type each)
- rows = samples / records
- **In ML language:**
  - feature columns → X (the inputs)
  - target column → y (what we predict)
- **Lab 01 starts from here — if these feel rusty, redo this in the demo notebook.**

POLICY

Vibe coding — welcome, with guardrails

You may use AI assistants (Colab AI, ChatGPT, Copilot…) everywhere in this course — labs included. ML is where they shine and where they lie.

✓ Great uses

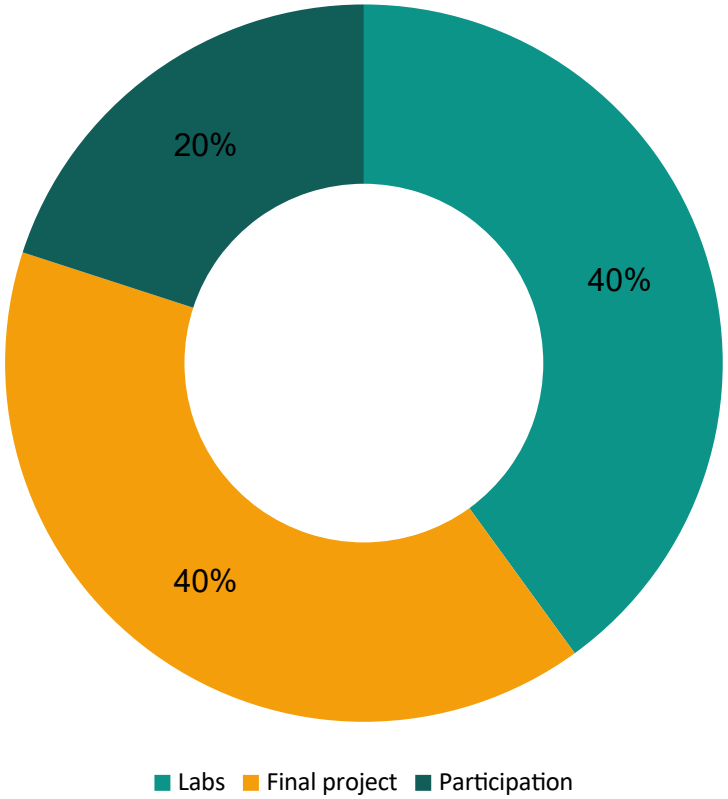
- Explain an error message
- “write me EDA code for this column”
- Boilerplate: train_test_split, plots
- “explain this metric like I'm five”
- Debug your pipeline step by step

✗ Where they fail you

- Invent plausible-but-wrong APIs
- Silently evaluate on the test set
- “improve” accuracy by leaking data
- Confident nonsense about metrics
- Code you can't explain = not yours

Rule of the course: you must be able to explain every line you submit. If you can't, ask the AI to explain it — then you own it.

How you'll be graded



- **Labs (40%) — 8 Thai-language labs in Colab**
  - graded on completion + honest attempts, not on max score
- **Final project (40%) — churn prediction capstone**
  - clean pipeline, honest evaluation, 5-minute presentation
- **Participation (20%)**
  - running demos along with me, asking questions

KEY TAKEAWAYS

1. 8 sessions: fundamentals (1–4) → model families (5–7) → capstone project (8).
2. Every session: English slides → run-along demo → Thai lab in Colab.
3. Toolkit (Colab, sklearn, pandas, seaborn) is pre-installed — zero setup.
4. AI assistants are allowed everywhere — but you must be able to explain every line you submit.

NEXT

Session 01 — Introduction to Machine Learning — your first model in 5 lines